

TASK 02 REPORT

INTERSHIP
CYBERSECURITY

20
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Week 2: Implementing Security Measures

1. Fixing Identified Vulnerabilities

In Week 2, I focused on resolving the vulnerabilities discovered during the initial assessment:

- **Input Validation:**
User input fields were validated to ensure they were not left blank and followed the correct format. This helped prevent injection attacks and input-based exploits.
- **Password Hashing:**
Instead of storing passwords in plain text, a hashing mechanism was applied using bcrypt. This added a layer of security by ensuring that even if the database is compromised, the actual passwords remain protected.

2. Enhancing Authentication

To protect access to sensitive features of the application, token-based authentication was implemented:

- After a successful login, a **JWT (JSON Web Token)** is generated and sent to the client.
- Middleware was added to verify this token for all important routes such as create, view list, and delete.
- Only valid token holders can access or perform CRUD operations, improving security against unauthorized access.

3. Securing Data Transmission

To protect data during transmission and prevent common attacks, security headers were added using **Helmet.js**:

- **XSS Protection:** Reduced the risk of cross-site scripting attacks.
- **Clickjacking Protection:** Prevented the site from being embedded in iframes on other websites.
- **MIME Sniffing Protection:** Ensured correct content types were interpreted by the browser.

Summary

By the end of Week 2, the security of the application was significantly enhanced through:

- Proper input validation
- Secure password handling
- Token-based authentication
- Implementation of important HTTP security headers

These improvements ensured that the web application followed essential security practices to guard against common vulnerabilities.